

RACES AND GAMES OF SKILL.

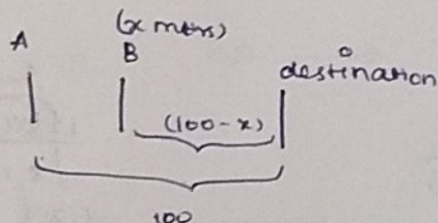
- 1). A gives B a start of 'x' meters.

Race $\rightarrow$ 100m

A has to run 100m

B has to run $(100-x)$ m

To win the race.



- 2). A gives B a start of 't' mins.

* A starts the race 't' min after B starts the race.

- 1) In a 100m race A covers the whole distance in 36s and B in 45s. By how many meters A beats B.

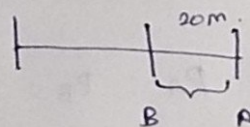
Ans: 20 meters

A Beats B by 20 mtrs.

100m $\leftarrow$ 45 sec

x $\leftarrow$ 9 sec (45-36)

$$\Rightarrow \frac{100}{45} \times 9 = 20.$$



- 2). In a 500m race A reaches the final point in 28s and B in 35s. By how much distance A beats B.

Ans: 100 mtrs

35s $\rightarrow$ 500

28s $\rightarrow$ x

$$\Rightarrow \frac{500}{35} \times 28 = 100 \text{ mtrs}$$

- 3). In a km race, A beats B by 100 meters. And B beats C by 150 mtrs in the same race. By how many mtrs does A beats C.

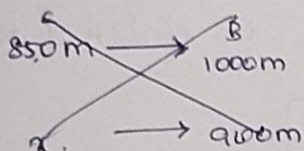
Ans: 235

$$\frac{A}{1000m} \rightarrow$$

$$\frac{B}{900m}$$

$$\frac{B}{1000m} \rightarrow$$

$$\frac{C}{850m}$$



$$\Rightarrow \frac{1000}{850} \times 900$$

B

C

$$1000m \leftarrow 850m$$

$$900m \leftarrow ?$$

$$\frac{850}{1000} \times 900 = 765m$$

A B C

1000m 900m 765m

$$\Rightarrow 1000 - 765 = 235m$$

A beats C by 235m.

4) In a race of 1000m, A can beat B by 100m. In a race of 400m B beats C by 40m. In a race of 500m, by how many meter A beats C.

Ans: 95m

A B C
1000m 900m 810m
500m 450m 405m

$$400m \rightarrow 360$$

$$900m \rightarrow x$$

$$\Rightarrow \frac{360}{400} \times 900$$

$$\Rightarrow 810$$

A beats C by 95m

5) In a 200m race A beats B by 31m, and C by 18m. In a race of 350m, by how many meters C beats B.

Ans: 25m

A B C
200m 169m 182m

C

B

182

169

350

x

$$\Rightarrow \frac{169}{182} \times \frac{175}{350}$$

$$= 325m$$

$$\Rightarrow 350 - 325 \Rightarrow c \text{ beats } b \text{ by } 25 \text{ m.}$$

b) In a 200m race, A ^{means B's data} beats B by 25m or 7 seconds. What is the time taken by A & B to finish the race.

Ans: 33 sec and 40 sec

$$\begin{array}{l} 25\text{m} \rightarrow 7\text{sec} \\ 200\text{m} \rightarrow x \end{array} \Rightarrow \frac{7}{25} \times 200 = 40 \text{ sec.}$$

$$A = 40 - 7\text{sec} = 33 \text{ sec}$$

(or)

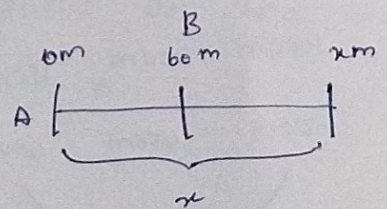
$$\begin{array}{l} 25\text{m} \rightarrow 7\text{sec} \\ 165\text{m} \rightarrow x \end{array} \Rightarrow \frac{7}{25} \times 165 = 33 \text{ sec}$$

$$B = 33 + 7 = 40 \text{ sec.}$$

$$\therefore (t_A)_{200} = (t_B)_{165} = 33 \text{ sec}$$

7. A runs 4 times as fast as B. If A gives B a start of 60m. How far must be the destination on the race course so that A & B reach the destination on the same time?

Ans: 80 meters



SXD.

$$S_A = 4S_B$$

$$\frac{S_A}{S_B} = \frac{D_A}{D_B} \Rightarrow A = \frac{x}{x-60}$$

$$\frac{S_A}{S_B} = 4$$

$$\Rightarrow 4x - 240 = x$$

$$\Rightarrow 3x = 240$$

$$\Rightarrow x = 80$$

$$D_A = xm$$

$$D_B = (x - 60) \text{ m.}$$

8. In a 500m race the ratio of speeds of A & B is 3:4. If A has a start of 140m. Then by how many meters A beats B.

Ans: 20m

start of
A $\rightarrow$ 140m

500m to win

A $\rightarrow$ 360m

3p $\rightarrow$ 360

B $\rightarrow$ 500m.

1p $\rightarrow$ 120m.

3:4
($\times 120$)

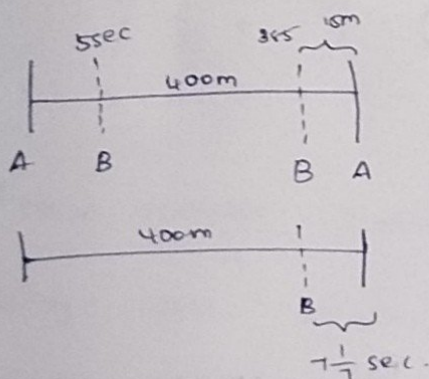
360:480

$\Rightarrow 500 - 480 \Rightarrow 20$ mtrs.

A beats B by 20m.

9. In a 400m race. A gives B a start of 5 sec and beats him by 15m. In another race of 400m A beats B by $7\frac{1}{7}$ sec. Find the speeds of A & B in m/s.

Ans: 8 m/s and 7 m/s



$7\frac{1}{7} - 5$ sec $\xrightarrow{B}$ 15m

$2\frac{1}{7}$ sec $\xrightarrow{B}$ 15m

$\frac{15}{7}$ sec $\rightarrow$ 7m.

(t_B) $385 = 55$ sec

(t_A) $400m = 50$ sec

$S_B = 7$ m/sec

$S_A = \frac{400m}{50sec}$

7m $\rightarrow$ 1 sec

385m $\rightarrow$?

$\Rightarrow \frac{1}{7} \times 385 =$

$S_A = 8$ m/sec.